

Data and AI in Economics

Payment Trajectories and Causal Delinquency:

Predicting Credit Card Default in Taiwan

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1. Why Predict Credit Card Default?



Research Question

Do earlier months of payment delinquency causally increase the probability of credit default beyond what the most recent payment behaviour already explains?



Motivation & Economic Relevance

Market impact: Credit card defaults cause significant losses to financial institutions and affect credit markets globally.

Risk management: Issuers must anticipate defaults. While payment history is predictive, correlation alone doesn't justify intervention.

Causal understanding: Does recent delinquency causally increase default risk, or is it only a symptom of deeper financial distress?

Customer segmentation: Are there distinct repayment trajectory archetypes that define high-risk groups?

2. Dataset Overview



UCI Dataset

- **Source:** Default of Credit Card Clients (Yeh, 2009)
- **Observations:** 30,000 credit card clients in Taiwan, 2005
- **Features:** 23 (demographics, credit limit, payment history)
- **Treatment:** PAY_0 (September Payment)
- **Target:** Binary default indicator (22% default rate)

Data Cleaning



Cleaned Dataset

- Unexplained feature values were removed.
- Payment status codes were standardized by collapsing redundant categories into a single label to simplify the ordinal scale.
- Rows after cleaning: 29,601 with a 22.313% overall default rate.

3. Three-Block Methodology

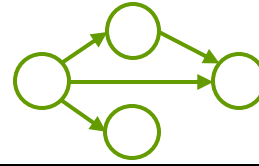


7a. Unsupervised

Purpose: Identify repayment patterns from payment history, independent of background.

Methods:

- **Clustering:** K-Modes (optimal for ordinal/discrete codes).
- **Validation:** Silhouette scores and hierarchical dendrograms.
- **Interpretation:** Heatmaps to visualize trajectory centroids.

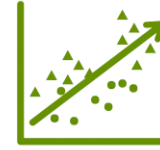


7b. Casual Inference

Purpose: Estimate the causal effect of September delinquency (PAY_0) on default probability.

Methods:

- **Causal Graph:** DAGs to map and control for confounders.
- **Estimation:** DoWhy library with propensity score weighting (ATE).
- **Validation:** Placebo treatment refuter.



7c. Supervised

Purpose: Predict defaults and identify key predictive features.

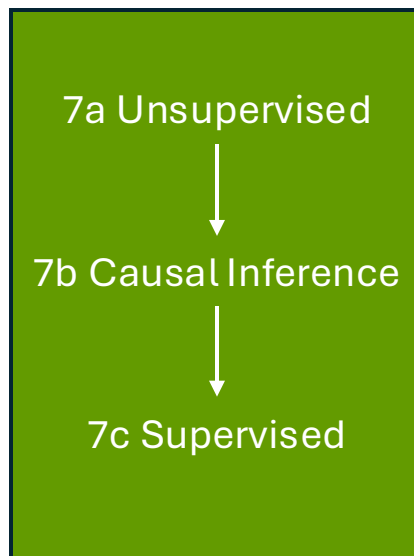
Methods:

- **Modeling:** Logistic Regression and Random Forest.
- **Evaluation:** AUC-ROC with 5-fold cross-validation.
- **Explainability:** SHAP values for feature importance.
- **Synthesis:** Cluster labels added as predictive features.

Synthesis (Section 8): Connect causal ATE, cluster labels, and SHAP importances to directly answer the research question.

4. Synthesis & Results

Methodology



Synthesis

- **Cluster labels(7a) are utilized as predictive features** within the supervised models in Section 7c.
- Convergent evidence is derived by **comparing SHAP feature importance** from the supervised block with the causal Average Treatment Effect.
- Evaluate if recent delinquency provides a **causal signal for default beyond historical patterns** while addressing Limitations.

Results

- **Unsupervised:** K-Modes grouped repayment patterns into distinct risk archetypes.
- **Causal:** DoWhy analysis signaled that recent delinquency causally increases default risk.
- **Supervised:** Random Forest outperformed logistic regression, with PAY_0 as the top predictor.
- **Cluster labels** offered minimal predictive value over raw historical data.

5. Limitations

Methodological Constraints

Cross-sectional snapshot: Data is from a single point in time, preventing proof of causal order.

No randomized intervention: Lacks a random experiment to properly isolate variable impacts.

Unstable backdoor assumption: Causal claims assume all hidden factors are accounted for, which cannot be proven.

Unmeasured confounding: Misses possible variables (e.g. Macroeconomic Shifts) that could skew results.

Generalisability

Time period: 2005 data may not reflect current credit markets.

Geography: Taiwan-specific patterns, transferability unknown.

Data Challenges

Class imbalance: 22% default rate, AUC-ROC preferred over accuracy.

Unspecified labels for the MARRIAGE and EDUCATION features → resolved by removing the affected rows.

References

- [1] Yeh, I. (2009). Default of Credit Card Clients [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C55S3H>.

Appendix

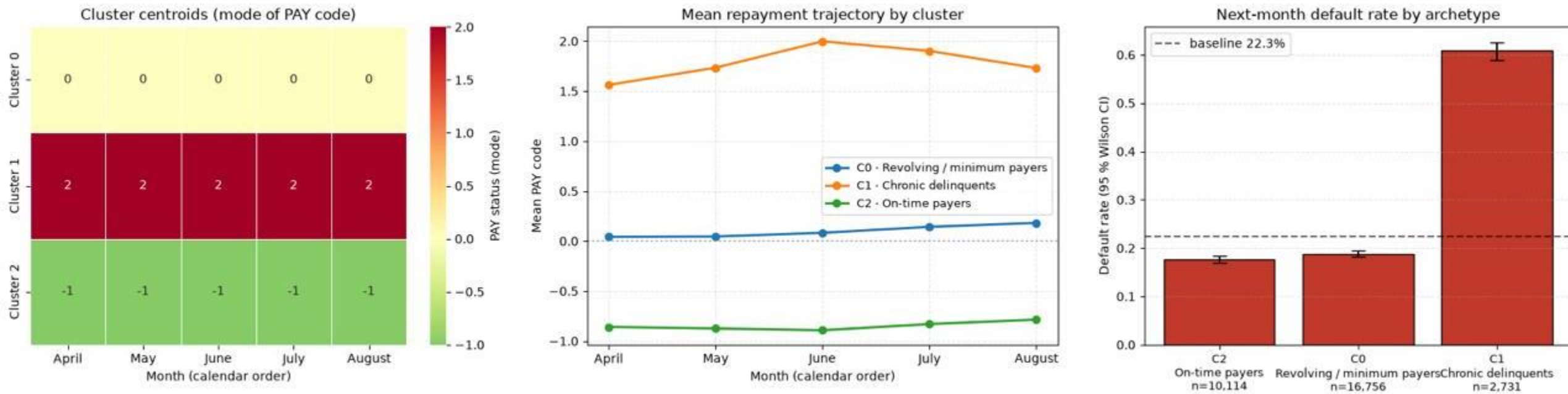


Fig. 1. Repayment Trajectory Archetypes Identified Using K-Modes Clustering.

Appendix

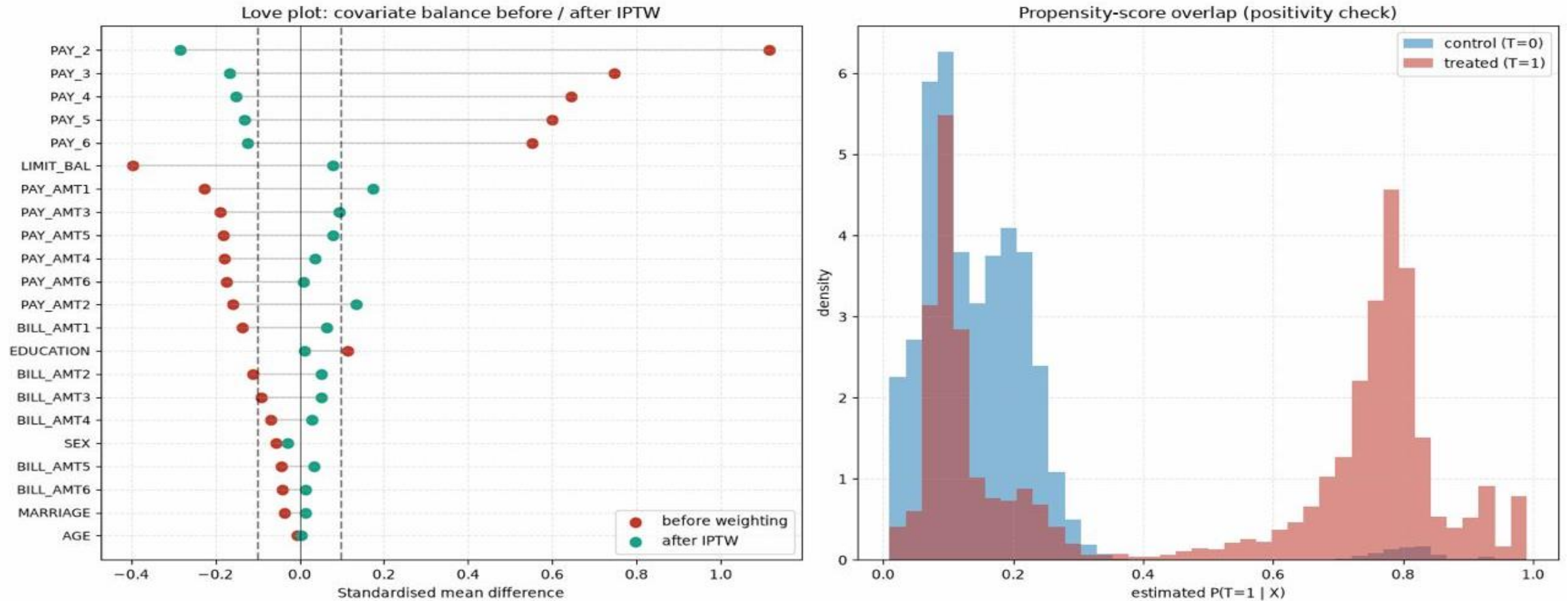


Fig. 2. Covariate Balance Diagnostics Before and After IPTW Weighting

Appendix

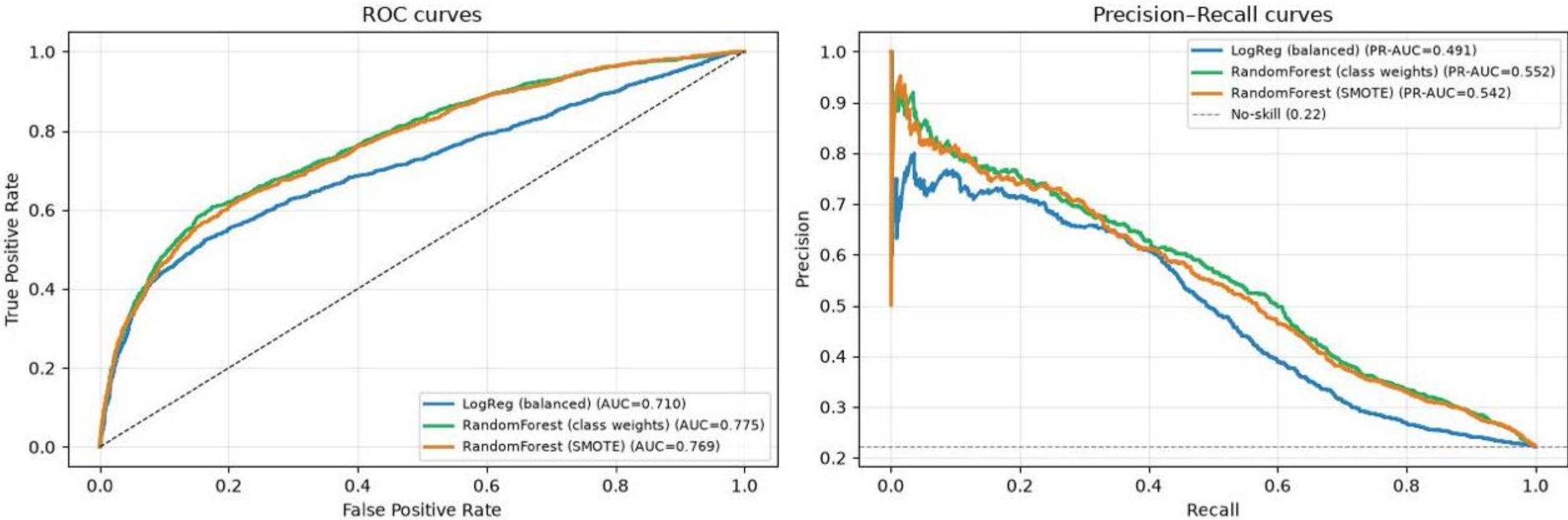


Fig. 3. Supervised Model Performance Evaluation Using ROC and Precision-Recall Curves